

R1 Color Fidelity | Final Evidence Casebook

A bounded, pixel-identified audit of pre-bake RGB outputs, runtime reproducibility and VAE isolation.

Frozen paired protocol

26 objects × No Adapter / GFL / LLH; same checkpoint, per-object seed and inputs; unique six-view targets.
Checkpoint SHA-256 0618d6b284ab47aa...; base runner SHA-256 e5c28e915ba137d3...; view order is recorded per row.
CIEDE/texture use identified PNGs; FG-PSNR/LPIPS/Edge-SSIM use the hash-linked Python 3.10 reloaded-RGB ledger.

Object-level paired effects (20,000 bootstrap resamples; seed 61008)

GFL – No Adapter	mean_fg_ciede2000		–19.654 95% CI [–26.370, –13.248] 24/26 lower
GFL – No Adapter	fg_psnr		+6.646 dB 95% CI [+5.556, +7.635] 25/26 higher
GFL – No Adapter	fg_lips		–0.0325 95% CI [–0.0536, –0.0107] 18/26 lower
LLH – GFL	mean_fg_ciede2000		+3.583 95% CI [+0.462, +6.712] 9/26 lower
LLH – GFL	fg_lips		–0.0129 95% CI [–0.0274, +0.0007] 13/26 lower
LLH – GFL	texture_fg_grad_mag		–0.0677 95% CI [–0.0813, –0.0549] 0/26 higher

Reading the evidence

The same two original Fig. 4 objects were visually tagged with a pink/purple cast in No Adapter, GFL and LLH outputs. GFL lowers average GT-referenced color error over 26 objects, but does not remove those two visible cases.

LLH has no reliable structure advantage over GFL and reduces measured texture/detail. Its average CIEDE2000 is worse; the small LPIPS decrease has a paired interval touching zero.

The color error exists in generated RGB before texture baking. A GT VAE round trip averages CIEDE2000 4.09 on the selected object, versus 22.03 for GFL; this rules against a simple VAE-only account, not every decoder interaction.

Runtime reproduction is environment-sensitive: three locked Fresh C examples reproduce the historical PNGs exactly in Python 3.13 and the earlier observer PNGs exactly in Python 3.10. The first captured value drift is FP16 scheduler latent scaling before UNet step 0.

Image identity

Each following panel is the exact PNG named in evidence_rgb/; no recoloring, GT adjustment, or per-image normalization was applied.
P1 raw images use the same six unique target views. Early denoising snapshots are shown only as trajectory context; they are not quality outputs.

Evidence rows: 78 paired RGB conditions; 22 paired effect summaries; 18 aligned late-stage rows.

Fig. 4 row 1 | persistent pink cast

Original object idx 4; seed 46; UID eac4b392b7b448a2b6ec77e8936abe1b; same six target IDs [14, 15, 0, 16, 12, 13].

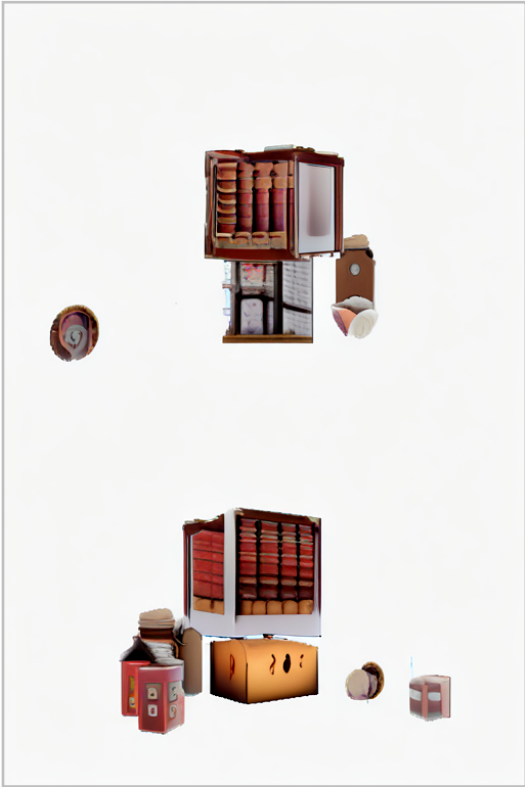
No Adapter: CIEDE2000 55.16; FG-PSNR 3.64 dB; FG-LPIPS 0.329
GFL: CIEDE2000 22.03; FG-PSNR 11.59 dB; FG-LPIPS 0.230
LLH: CIEDE2000 32.26; FG-PSNR 8.97 dB; FG-LPIPS 0.250

GT



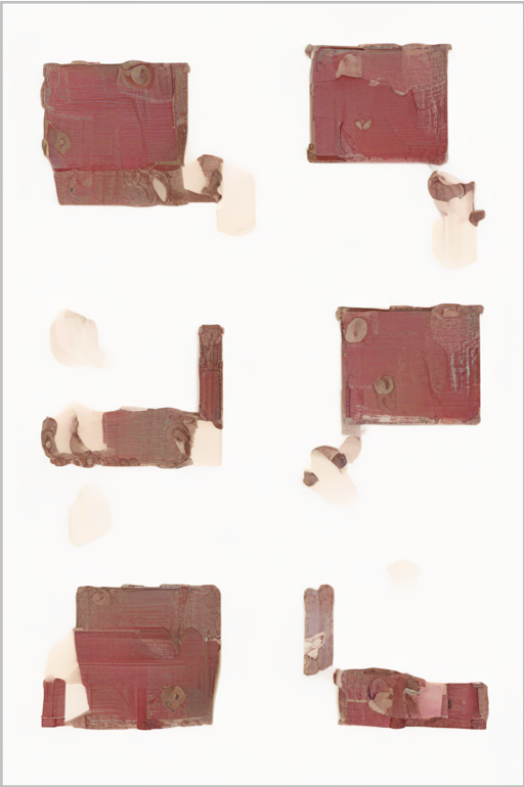
635c6001dff80148

No Adapter



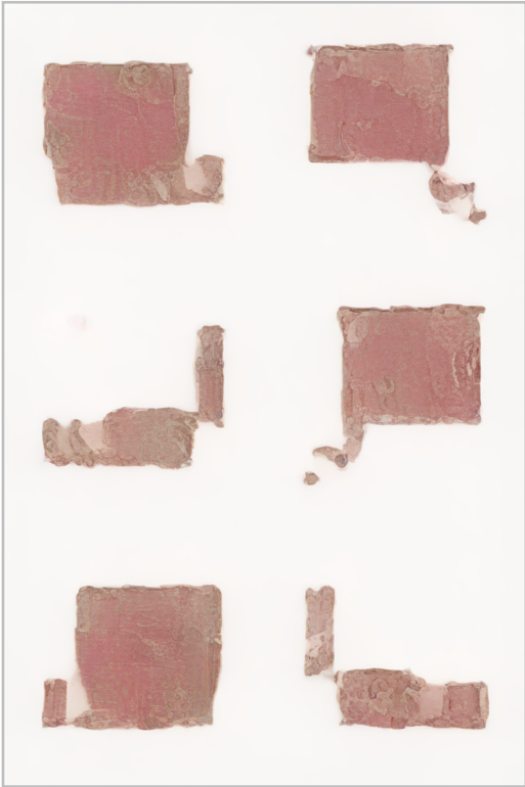
b8efd3c0a60f389e

GFL



2d6f3273dc3600f0

LLH



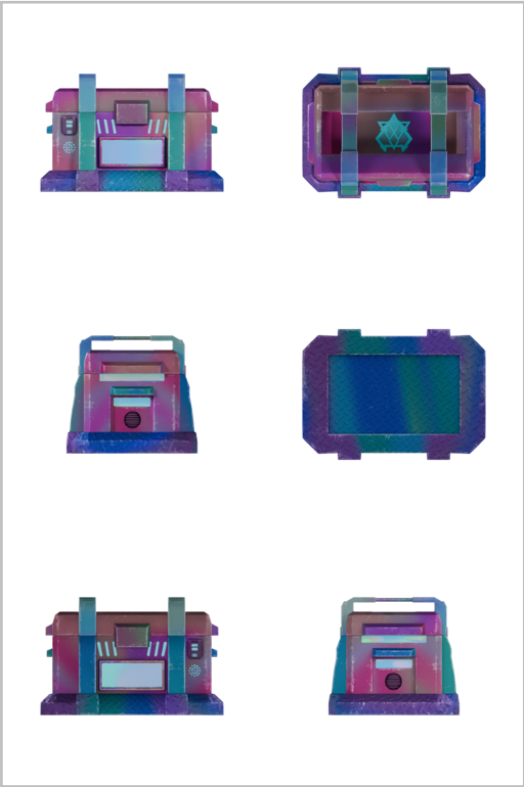
4dfe29eafb1e5d35

Fig. 4 row 3 | persistent hue error

Original object idx 12; seed 54; UID ed600cc9d91448939edeeb3a6d858448; same six target IDs [14, 15, 0, 16, 12, 13].

No Adapter: CIEDE2000 49.36; FG-PSNR 4.54 dB; FG-LPIPS 0.309
GFL: CIEDE2000 25.76; FG-PSNR 11.81 dB; FG-LPIPS 0.274
LLH: CIEDE2000 33.24; FG-PSNR 9.77 dB; FG-LPIPS 0.275

GT



f5b9cef6cc4f3bcd

No Adapter



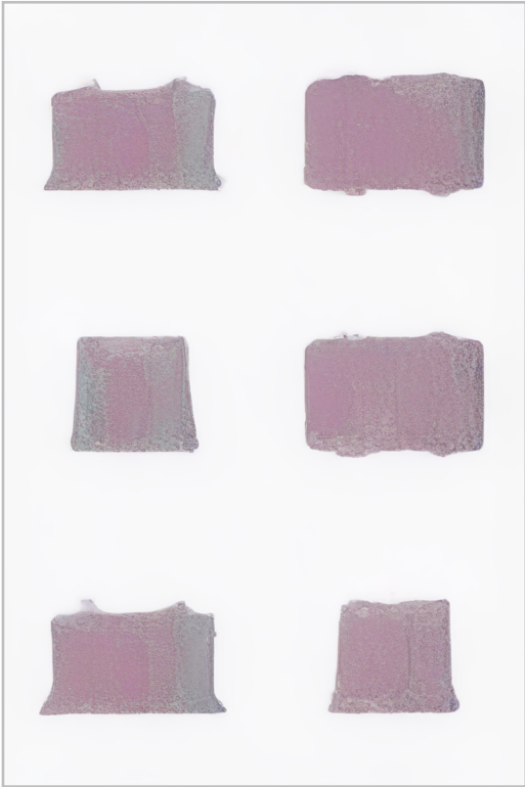
9df4278a773f18bd

GFL



33cb0cadd8b644e6

LLH



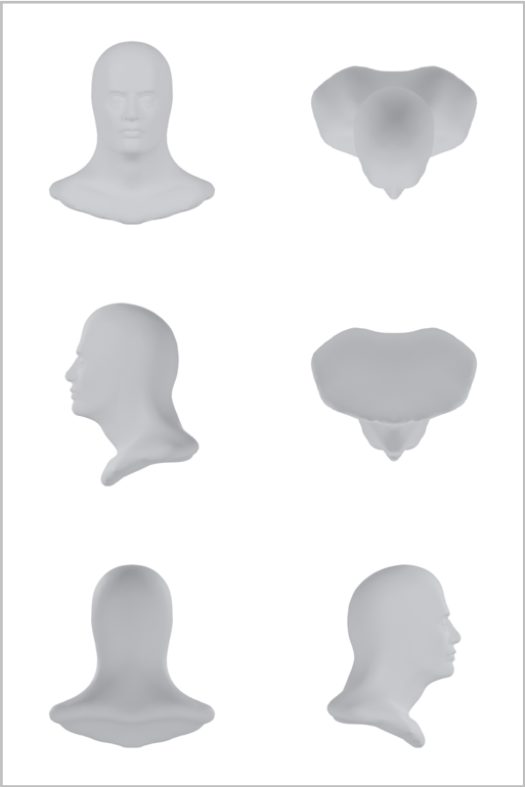
7310bc3476302434

Fresh C | typical color / detail control

Fresh C UID ca887928bb664bcba121219f0af41293; no visual pink cast or fine-detail-loss tag in the frozen review.

No Adapter: CIEDE2000 13.76; FG-PSNR 12.71 dB; FG-LPIPS 0.094
GFL: CIEDE2000 8.22; FG-PSNR 18.11 dB; FG-LPIPS 0.116
LLH: CIEDE2000 3.88; FG-PSNR 23.88 dB; FG-LPIPS 0.068

GT



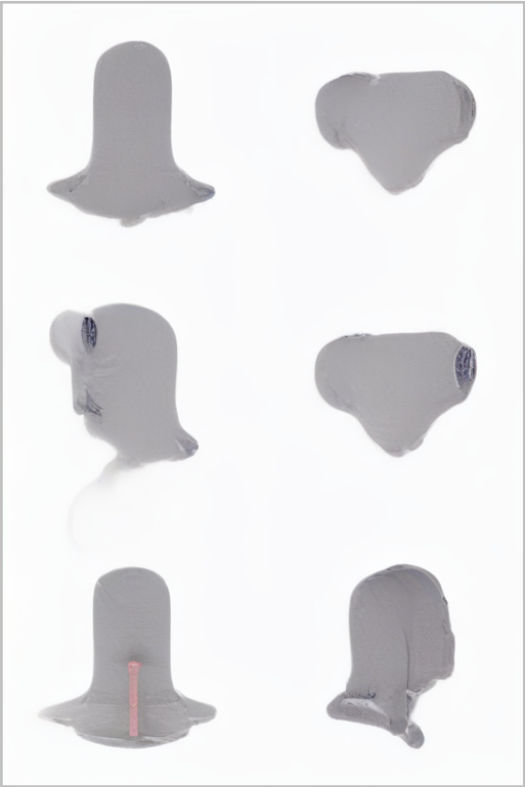
295e32f0384146c4

No Adapter



8fe6c9aad3f683d3

GFL



ddfe23009598b4d8

LLH



70d023472288c8ed

Fresh C | detail-loss control

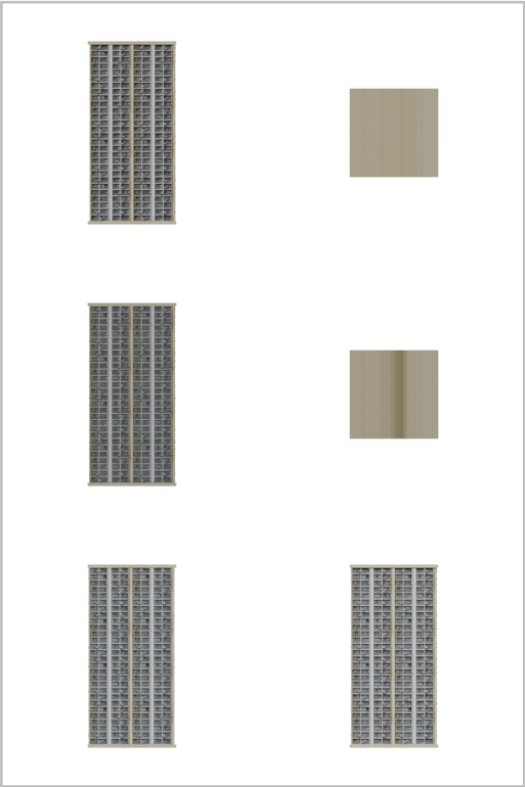
Fresh C UID c76ac44df995482185077da81939b306; fine-detail loss tagged; no visual pink cast. Also used in P0 runtime identity audit.

No Adapter: CIEDE2000 31.23; FG-PSNR 6.37 dB; FG-LPIPS 0.279

GFL: CIEDE2000 13.52; FG-PSNR 16.22 dB; FG-LPIPS 0.177

LLH: CIEDE2000 15.19; FG-PSNR 14.12 dB; FG-LPIPS 0.208

GT



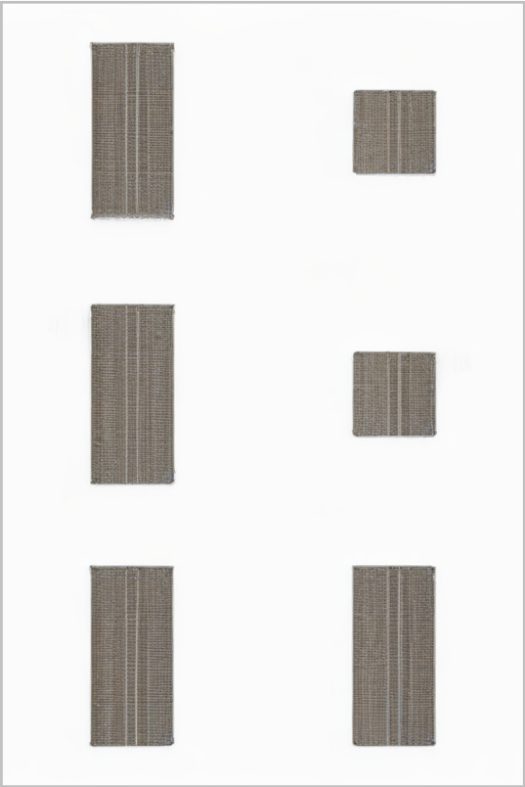
b9e39632a11d83b1

No Adapter



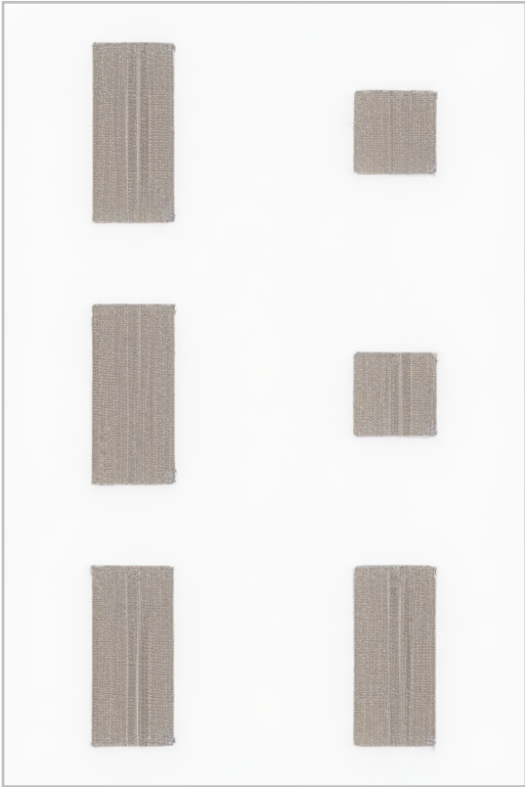
a5857fa5bd6c3790

GFL



773b21bd82e52236

LLH



225af3d596855ed7

P0 | Runtime identity — freshc_panel_01

UID c76ac44df995482185077da81939b306; same locked GFL checkpoint, object seed, input tensors, config and unique-six target views.

Archive ↔ Python 3.13.5: byte-identical; the repeat trace is also exact.

Archive ↔ Python 3.10.20: MAE 8.76/255; changed-pixel fraction 0.9998.

First captured cross-runtime value drift is the FP16 latent × scheduler sigma before UNet step 0.

Historical archive



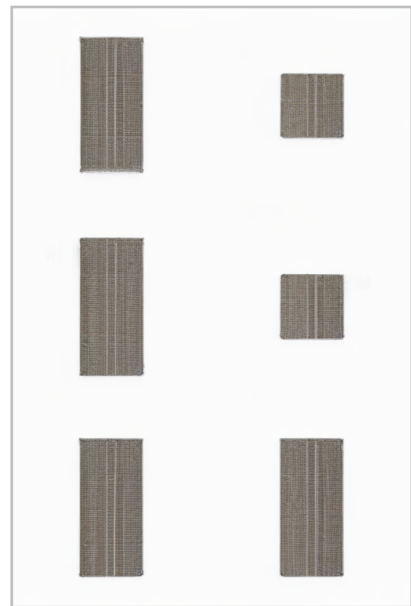
8a7d5c30df97abd0

Python 3.13.5



8a7d5c30df97abd0

Python 3.10.20



773b21bd82e52236

P0 | Runtime identity — freshc_panel_03

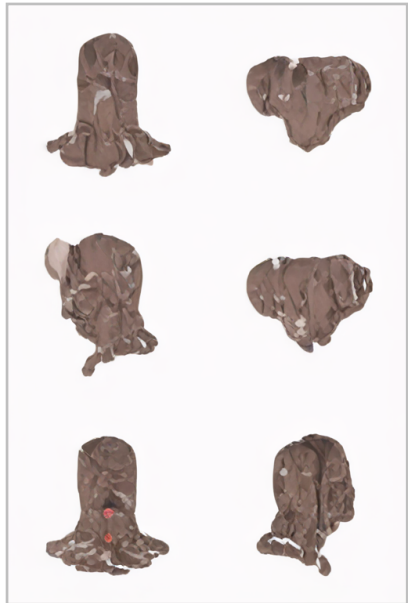
UID ca887928bb664bcba121219f0af41293; same locked GFL checkpoint, object seed, input tensors, config and unique-six target views.

Archive ↔ Python 3.13.5: byte-identical; the repeat trace is also exact.

Archive ↔ Python 3.10.20: MAE 13.17/255; changed-pixel fraction 0.9997.

First captured cross-runtime value drift is the FP16 latent × scheduler sigma before UNet step 0.

Historical archive



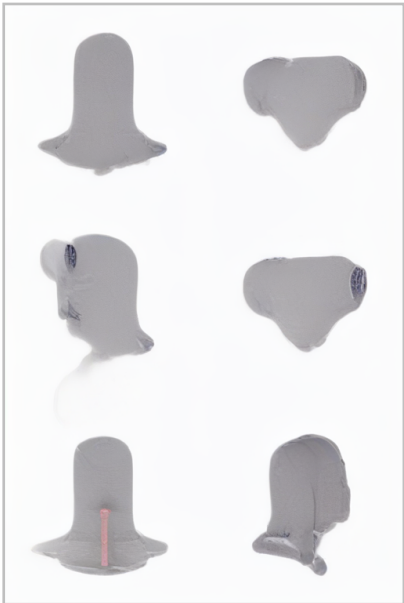
b73bab6acd628d11

Python 3.13.5



b73bab6acd628d11

Python 3.10.20



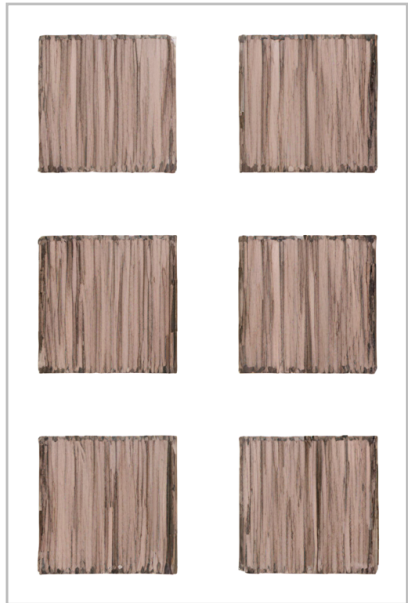
ddfe23009598b4d8

P0 | Runtime identity — freshc_panel_20

UID 06fa4974f8cd4d4ea792e98fa1521457; same locked GFL checkpoint, object seed, input tensors, config and unique-six target views.

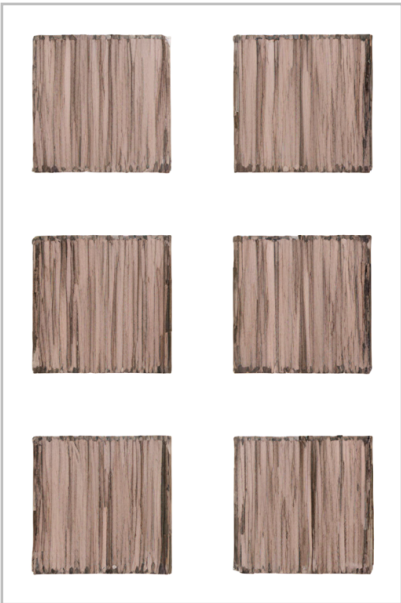
Archive ↔ Python 3.13.5: byte-identical; the repeat trace is also exact.
Archive ↔ Python 3.10.20: MAE 12.65/255; changed-pixel fraction 0.5147.
First captured cross-runtime value drift is the FP16 latent × scheduler sigma before UNet step 0.

Historical archive



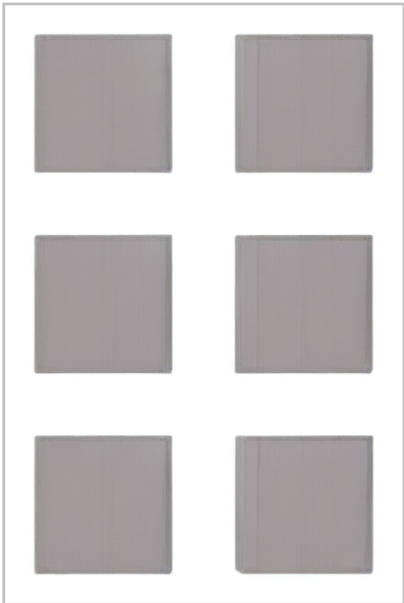
a123c75f7feaadcd

Python 3.13.5



a123c75f7feaadcd

Python 3.10.20



3f1c140d39e76eef

P1 | Color trajectory in one aligned pink case

Fig. 4 row 1 GFL; Python 3.10 output SHA matches frozen P1 RGB exactly. Intermediate images are decoded latent snapshots.

After update 25



6e88d59132f1cc7a

CIEDE 42.3; Δa^* -11.1; Δb^* -5.1

After update 33



3ddb39318c333ef

CIEDE 40.2; Δa^* -5.2; Δb^* -3.6

After update 40



676e2d692c45b885

CIEDE 37.4; Δa^* +7.9; Δb^* -2.0

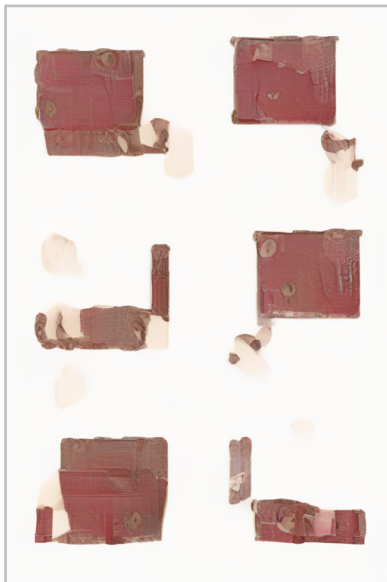
After update 45



2d03f9c274bbadb

CIEDE 30.3; Δa^* +10.5; Δb^* +1.1

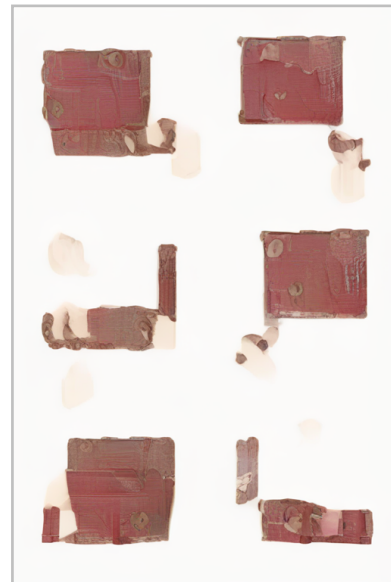
After update 49



2d6f3273dc3600f0

CIEDE 22.0; Δa^* +2.0; Δb^* -0.6

After update 49 / final



2d6f3273dc3600f0

CIEDE 22.0; Δa^* +2.0; Δb^* -0.6

The object is not interpretable in early high-noise snapshots. Magenta surfaces are visible by update 40 in this case.
This sampled trajectory localizes the observed pattern to late denoising, not the exact first latent step.

P1 | VAE encode-decode isolation

Same Fig. 4 row 1 GT input; Python 3.10.20, torch 2.7.0+cu128, diffusers 0.20.2; normalization recorded in the metrics JSON.

GT round trip mean CIEDE2000: posterior mode 4.086; fixed-seed sample 4.087.

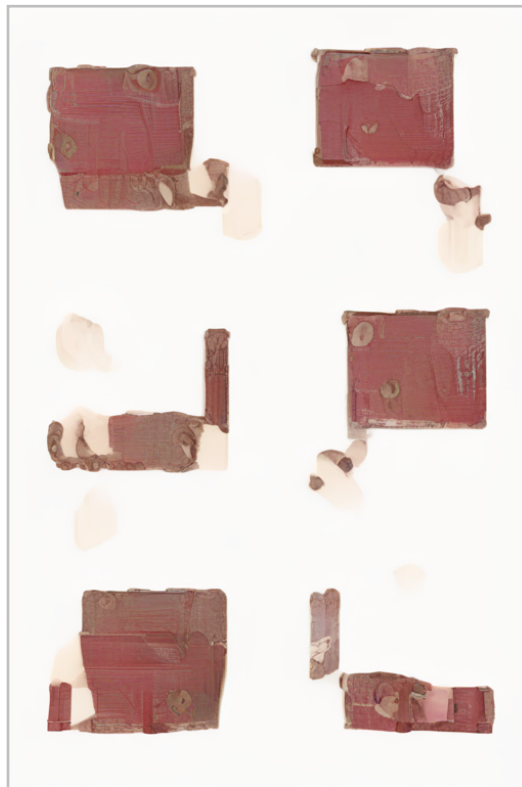
Compare with the aligned GFL output: CIEDE2000 22.032, Δa^* +1.977; the round trip's mean Δa^* is about -1.35 .

GT input



635c6001dff80148

GFL final



2d6f3273dc3600f0

VAE round trip · posterior mode



c1cff802d0f0d55

VAE round trip · sample seed 42



d1e26f21f98bb260